Aftershock

Building Safety Coordinator · Kestrel Bay

10 DAYS · 41 STOPS · GRADE 6 · AFTERSHOCK_MS

THE OPENING CARD

An earthquake hit Kestrel Bay three days ago. The town is still standing, unevenly. You are the building safety coordinator, which means you say which buildings people may go back into. Green to enter. Yellow for part of it. Red for nobody. Four hundred families are sleeping in halls tonight, waiting on what you decide, and the hospital has a yellow card on its front door. The shaking is not finished either. Upper Town sits on granite and lost its chimneys. The Flats sit on made ground and lost their streets, and four hundred families are waiting on you to say which of them can go home.

DAY 1 OF 10

Two towns, one earthquake

BEFORE THE DAY — GREET

Two teams in one office, and nobody has been introduced

Half the office is from the council and half arrived on Saturday. By Friday somebody has to trust somebody else's check. Put a name to as many of them as you can now, because a signature you cannot picture is one you will check twice.

PLAN CARD 77W

Friday, three days after, and nothing about it makes sense yet. Upper Town sits on granite. It lost its chimneys and kept its buildings. The Flats are eleven hundred metres away, on the same fault, with sand in the gutters and a six-storey block leaning over. Marisol Okonkwo will not sign an assessment plan until somebody says why. Today you read the two records side by side, and decide which half of town gets the assessors first.

BRIEFING 18W

Upper Town lost its chimneys. The Flats lost its streets. Both are the same distance from the source.

WHAT YOU DECIDE 16W

Work out why the same shaking wrecked one half of Kestrel Bay and spared the other.

1.1 How far away it started

SEISMIC NETWORK · A ROOM · BALLPARK

SCENE 31W

The vault trace shows the first wave at 04:12:19 and the strong shaking at 04:12:32. Inês Cardoso, the network technician, wants a distance before anybody draws a circle on a map.

GUIDE 57W

Five numbers, and two belong to the earthquake and not to this station. The magnitude, and how deep it started. Two more are clock times, and only the gap between them carries distance. Ask of each what it measures. And note what one station gives you. A distance, not a place. The circle still needs other stations.

ASK 6W

How far away did it start?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the first wave travels faster than the strong shaking that follows it

VERDICT 72W

Both waves leave at the same moment, and one of them travels faster. So the gap between them grows the further you are from the source. Close in they arrive almost together. A hundred kilometres away there are thirteen seconds between them. That gap gives a distance, and a distance on its own is a circle round the station. Three stations give three circles, and where they cross is where it started.

TAKEAWAY 20W

The gap between the two arrivals grows with distance, so one station gives a distance and three give a place.

1.2 Ground that people put there

GEOTECHNICAL · A PERSON · PROTOCOL

SCENE 37W

Elena Navarro, the ground engineer, has an 1892 survey and a photograph from 1948. Where Bay Road runs now there was a tidal creek. The port was pumped into place out of a dredger over four summers.

GUIDE 63W

Four kinds of ground on the left. Four ways ground behaves on the right. Pair them by asking two things. How loose is it? Can the water get out? Age is not the clue it looks like. One of these is the loosest and wettest of all. It is a line on an old survey, and you cannot see it from the street.

ASK 10W

Say what the ground under the Flats is made of

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

ground can be natural, or it can be put there by people · saturated means every gap between the grains is already full of water

VERDICT 80W

Fill that is pumped in through water lands loose, and it stays full of water. Shake it and the grains try to shuffle closer together. The water in the gaps cannot get out fast enough, so the water takes the load instead of the grains. Grains that are not pressing on each other cannot hold anything up, and the ground behaves like a thick liquid. Old ground has had a hundred centuries to pack down. This has had eighty years.

TAKEAWAY 12W

Loose, wet, young ground behaves in a way old rock does not.

1.3 Where the first day of assessment goes

STRUCTURAL ASSESSMENT · A ROOM · ALLOCATE

SCENE 34W

Six assessors are free for one day. Four hundred families are waiting on a decision in the Flats. Marina Court and the Parade are already fenced off, and Upper Town is mostly lived in.

GUIDE 65W

Six assessor-days is all you have, and each package you drag in spends some. Watch the list of questions rather than the buildings: it shows which ones your plan can answer. Every package you buy is a question you can answer and one you have given up. The fences already up cost nothing. Commit when the answers you can give are the ones today needs.

ASK 18W

Can your plan still get the waiting Flats families a decision today, knowing some questions have to wait?

BEHIND THE BUTTON 142W

Why the pool is the whole problem. Six assessors for one day is a fixed amount of attention, and the Flats alone could use all of it. The question is not what is worth checking — nearly everything is. It is which checks change what happens before tonight.

What makes an answer worth buying today. Four hundred families are waiting to hear whether they can go home. A check that can answer that by this evening is worth more than a better answer next week, because the delay has already made the decision for them.

Why the fences are on the board at zero. They are already up. They cost nothing to keep and they answer nothing new. They are there to be noticed: a plan that spends days re-checking what is already closed has bought a question it had already answered.

TAKES AS READ

an inspection queue is a decision, not a list

VERDICT 68W

There are six assessor-days and more good work than that. Every inspection bought is another one given up. The Flats sweep can send four hundred families home tonight, or keep them out on purpose. Marina Court and the Parade have fences doing that job already, so more detail there changes little today. Upper Town is useful and not urgent. Buy the answer that can still change what happens.

TAKEAWAY 14W

Spend a team on the questions whose answers can still change what happens today.

END OF THE DAY 18W

One earthquake produces as many shakings as there are grounds to shake, and the ground is the difference.

DAY 2 OF 10

The number everybody has already heard

BEFORE THE DAY — TRIAL

Walk Upper Town before you place a single notice

Three days after the earthquake, and you have not been here before. Walk the near buildings once on foot. Nothing gets a notice from a map, and the streets on the plan card mean nothing until you have seen which way the ground falls.

PLAN CARD 69W

Saturday, and the town has two numbers for the same earthquake. One agency said 6.6 within twenty minutes. Another said 6.8 six hours later. Rei Tanaka can explain the difference to an engineer, and has spent the morning failing to explain it to a radio station. Funmi Adeyemi has to put something in a notice by four. Today you say what the magnitude is, and what it is not.

BRIEFING 14W

Two agencies published two magnitudes, and the difference is being read as a disagreement.

WHAT YOU DECIDE 14W

Pin the magnitude and the location down, and separate them from what people felt.

2.1 The same event, recorded twice

SEISMIC NETWORK · A ROOM · BALLPARK

SCENE 33W

Inês Cardoso has two records of the same earthquake, from stations eleven hundred metres apart. One trace is short and sharp. The other is three times taller and runs half a minute longer.

GUIDE 56W

The same wave reached both stations. Work out how fast it travelled, from how far apart they are and how much later it arrived at the second one. Speed is the distance divided by the time. One tile is how much taller the second trace is, which is a different question about the same two records.

ASK 10W

How fast did the wave travel between the two stations?

BEHIND THE BUTTON 126W

Why the speed is worth knowing. It tells you what the wave travelled through. A wave through solid rock runs fast, around three or four kilometres a second. The same wave slows right down in loose wet ground, and that slowing is part of why it grows.

Why the two traces look so different. The far station stands on the Flats, where the ground is soft. Soft ground shakes further and for longer, so the trace is taller and runs on after the rock has gone quiet. Neither instrument is broken.

What each record is. A measurement of the shaking where that instrument stands. It is not a measurement of the earthquake, which is why two records of one earthquake can disagree and both be right.

TAKES AS READ

an instrument records the shaking where it stands, not the earthquake itself

VERDICT 141W

Both instruments caught the same wave. The far one caught it 0.4 of a second after the near one. Speed is distance ÷ time, so 1,100 ÷ 0.4 = 2,750 metres a second. That is fast, and it tells you what the wave came through: solid rock carries a wave at thousands of metres a second, and loose wet ground carries it far more slowly. The wave spent most of its journey in rock. The rest of the difference between the two records is about where the instruments stand. The far station stands on soft ground, so its trace is taller and it runs on for longer. Neither instrument is wrong. Each one recorded the shaking in its own place, which is what an instrument does, and it is why two records of one earthquake can disagree and both be right.

TAKEAWAY 17W

A record describes one place, so two records of one earthquake can disagree and both be right.

2.2 Why the number moved

HAZARD & FORECASTING · INCIDENT BASE · A PERSON · CHOICE

SCENE 25W

Tanaka has both bulletins on the desk. The early one used the first stations to report. The later one used the whole network, hours afterwards.

GUIDE 53W

All four options explain a number that moved. They differ in what changed. The earthquake? The data? The sums? The scale? Only one of those can change six hours later. Ask what the second bulletin had that the first did not. Get it wrong and the town hears that the earthquake grew overnight.

ASK 18W

The second figure used more stations than the first. What got bigger — the earthquake, or the measurement?

BEHIND THE BUTTON 169W

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Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a magnitude is worked out from records, so more records can change it

VERDICT 78W

A magnitude is not read off the ground. It is worked out from what the instruments recorded, so it gets better as more of them report. The first bulletin used the handful of stations that had answered in the first minutes. The later one used the whole network. Going from 6.6 to 6.8 sounds tiny, and it is not. That step means about twice the energy. The earthquake never changed. How much of it had been measured did.

TAKEAWAY 13W

A number that changes means the measurement improved. The event did not change.

2.3 A notice somebody acts on

PUBLIC SAFETY · A ROOM · SEQUENCE

SCENE 22W

Adeyemi has two hundred words, and a town that has heard two different numbers and a rumour about a bigger one coming.

GUIDE 60W

All four of these go in the notice. Nothing is being left out. Ask which one somebody can act on while standing in the street. Then ask which is a fact they will repeat later. Two hundred words are read in the order they are printed. A number going from 6.6 to 6.8 changes nothing about what to do now.

ASK 9W

Say it in a way that survives being repeated

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a public notice is read once, quickly, by somebody who is tired

VERDICT 71W

People act on the first thing they can use, so the instruction goes at the top. The reason comes next, because a reason is what makes an instruction survive being passed on. The revised number matters to scientists and changes nothing about what to do outside a building. The chance of aftershocks goes last, as a rate. As a rate it stays a chance. As a date it becomes a promise.

TAKEAWAY 18W

A notice earns its place by changing what somebody does, and everything else in it competes with that.

END OF THE DAY 16W

A magnitude is a measurement with a method attached, and revising it is the method working.

The building that looks worst

BEFORE THE DAY — FOLLOW

Mbeki walks the line where the ground changes

The line between the hard rock and the soft fill is not on any map closely enough to use. Mbeki, the geotechnical field technician, can see it in the ground. She walks it once this morning. Stay with her, and watch where she stops.

PLAN CARD 72W

Monday, and Marina Court is on the front page. Six storeys of flats leaning over the marina, photographed from the water. That photograph has frightened the town more than anything the office has said. Elena Navarro has been under it. The slab is uncracked and the frame is straight. Okonkwo will not sign the demolition of a sound building on the strength of a photograph. Today you work out what actually failed.

BRIEFING 19W

Marina Court is leaning eight degrees and its structure is undamaged. The queue outside it is not about engineering.

WHAT YOU DECIDE 12W

Separate what frightens people from what is actually holding the building up.

3.1 The slab is fine and the building is not

GEOTECHNICAL · A ROOM · CHOICE

SCENE 35W

Navarro's survey: the slab is whole and level within itself, and the whole block has tipped. One side is down three hundred and forty millimetres, the other up ninety. Sand fans on the low side.

GUIDE 57W

Four options, and the survey has to be explained whole. The slab is not broken. It is level within itself. One side is down and the other is up. Sand came up on the low side. Ask of each option how many of those it covers. A frame that broke would bend, not tip in one piece.

ASK 19W

The slab is whole and level within itself, and the whole block has tipped. Where did the movement happen?

BEHIND THE BUTTON 169W

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TAKES AS READ

a foundation spreads a building's weight out into the ground

VERDICT 73W

Everything here points underneath the building rather than inside it. The slab is whole. The frame is straight. The whole block has tipped like a tray, one edge down and one edge up, with fresh sand pushed up on the low side. That is ground that stopped holding weight. Shake wet fill hard enough and the water takes the load, the grains stop pressing on each other, and one side sinks into it.

TAKEAWAY 10W

A building can fail without anything in the building breaking.

3.2 What eight degrees does

STRUCTURAL ASSESSMENT · A PERSON · CHOICE

SCENE 29W

Okonkwo has the column loads out. At eight degrees the weight above each column no longer lands over the middle of it. The gap grows the higher you go.

GUIDE 56W

All four options say something true about a leaning building. They differ in whether they name a force or a feeling. Ask of each whether an engineer could put a number on it. A column carries weight best straight down its middle. Eight degrees moves the weight off the middle, and the gap grows higher up.

ASK 10W

What is the engineering objection to leaving Marina Court standing?

BEHIND THE BUTTON 169W

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TAKES AS READ

a column is built to be pushed straight down, along its own length

VERDICT 77W

A column is strongest when the weight lands straight down its middle. Lean the building and that weight arrives off to one side, so the column is bent as well as squashed. Bending is the job it was not built for, and the offset grows the further up the building you go. The question is not how frightening eight degrees looks. It is whether those columns have enough left in them for a load path nobody designed.

TAKEAWAY 18W

A column carrying its load off to one side is doing a job it was not built for.

3.3 What knocking it down would fix

PUBLIC SAFETY · A ROOM · VALUE

SCENE 30W

The mayor's office wants Marina Court pulled down this week. Delacroix has twelve blocks next to it on the same made ground, and five days before the fence is reviewed.

GUIDE 59W

Five days, four proposals, and somebody wanting a demolition this week. Open each proposal to see what decision its result could change. Marina Court is already fenced off, so evidence about that one building changes nothing. The twelve blocks next to it on the same made ground are where the open decision is. Commit the days you would spend.

ASK 13W

Which investigation buys something that helps beyond the one building already fenced off?

BEHIND THE BUTTON 138W

What is already decided. Nobody is going back into Marina Court this week whatever a test says. The open question is whether the ground that damaged it also needs action on the twelve blocks around it — and that is a question about the ground, not about the building.

Why the worst damage is the trap. Instrumenting the most damaged building gives dramatic results and answers a question that is already closed. It is the most sensible-sounding way to spend five days and learn nothing that changes anything.

What made ground is. The Flats were built on fill — soil brought in to make the land usable. Fill shakes differently from natural ground, and all thirteen buildings sit on the same fill. So one campaign of ground tests bears on twelve open decisions instead of one closed one.

TAKES AS READ

a decision can treat what you can see, or what caused it

VERDICT 66W

Pulling Marina Court down removes the building everybody can see. It says nothing about the ground under the twelve blocks on the same fill. Digging and testing that ground does. The fence already holds the hazard everybody knows about. So the five days are better spent on the question nobody has answered. More readings on one building answer a smaller question than the town is asking.

TAKEAWAY 16W

The best measurement is the one that could change a decision you have not made yet.

3.4 How far away it started

SEISMIC NETWORK · A ROOM · CALLBACK · BALLPARK

SCENE 31W

The vault trace shows the first wave at 04:12:19 and the strong shaking at 04:12:32. Inês Cardoso, the network technician, wants a distance before anybody draws a circle on a map.

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Five numbers, and two belong to the earthquake and not to this station. The magnitude, and how deep it started. Two more are clock times, and only the gap between them carries distance. Ask of each what it measures. And note what one station gives you. A distance, not a place. The circle still needs other stations.

ASK 6W

How far away did it start?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

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Both waves leave at the same moment, and one of them travels faster. So the gap between them grows the further you are from the source. Close in they arrive almost together. A hundred kilometres away there are thirteen seconds between them. That gap gives a distance, and a distance on its own is a circle round the station. Three stations give three circles, and where they cross is where it started.

TAKEAWAY 20W

The gap between the two arrivals grows with distance, so one station gives a distance and three give a place.

END OF THE DAY 18W

A building can fail without breaking, and the thing that failed may be the ground it stands on.

DAY 4 OF 10

The one everybody assumed was fine

PLAN CARD 76W

Tuesday, and the school is the quiet problem. Bay Road School came through with a green placard, and the district would like four hundred children back next week. Alan Whitcombe placarded it from outside, and wrote that the gymnasium was locked. The gym is a single-storey hall with a long roof and heavy panels for walls. Today you work out what such a hall does in an earthquake, and whether the children go in on Monday.

BRIEFING 17W

Bay Road School is green, term starts in a week, and nobody has been inside the gym.

WHAT YOU DECIDE 14W

Look at a green building properly, and find what a walk-past could never see.

4.1 A hall with nothing in the middle

STRUCTURAL ASSESSMENT · A ROOM · CHAIN

SCENE 31W

The gym at Bay Road School has a long roof, heavy wall panels and nothing bracing it in the middle. One panel fixing tested badly. Everything big in there looks fine.

GUIDE 61W

Put the handovers in the order the sideways push really travels, from the moving panel to the ground. Then name the one required handover that would break the path if it failed. The biggest part is not the answer. A path is only as good as its weakest handover, and the readings on each link tell you which one that is.

ASK 10W

Which link decides whether the panel's push reaches the ground?

BEHIND THE BUTTON 166W

What is being handed on. In an earthquake the ground moves and the building's mass wants to stay put. That push has to be carried by something, passed along, and finally delivered to the foundation. Each pass is a fixing or a member. If one of them cannot carry its share, the path stops there.

Why a gym is the hard case. A long roof with heavy wall panels and nothing bracing the middle has very few paths available. The roof has to act as one big flat plate and hand the push to the end walls, and the panels have to be tied into that plate. Lose one tie and the panel is on its own.

Why undamaged parts are not comfort. The beams and columns can be perfect while the fixings between them are not, and a fixing is the thing the drawings say least about. One panel fixing already tested badly, which says something about that kind of fixing everywhere in the building.

TAKES AS READ

a sideways push has to travel through a building to reach the ground

VERDICT 76W

A shaking building throws its own weight sideways, and that push has to get to the ground. From the panel it goes into the fixing at the top. Then into the roof, along to a wall that can take it, and down. Every step has to hold. The panel is heavy and the roof is big, and neither matters if the small fixing between them lets go. Size is not strength when a link is missing.

TAKEAWAY 17W

A load path works only if every link in it carries the force onward to the ground.

4.2 What the tested fixings actually carried

MATERIALS & TESTING · A ROOM · BALLPARK

SCENE 30W

Kirsten Sørensen has pulled two fixings out of a spare panel in the yard. The drawing says forty kilonewtons each. The first came out at twenty-four, the second at twenty-seven.

GUIDE 57W

Five numbers, and two belong to other questions. The roof span, and the number of panels. One is what the drawing promised, not what the wall got. Ask of each whether it is what the fixings carried or what the wall asks of them. And note the limit. Two pulls is not enough to call anything safe.

ASK 13W

How does what the fixings carried compare with what is asked of them?

BEHIND THE BUTTON 106W

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Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a number on a drawing is a claim, and a test is a check on it

VERDICT 75W

Two fixings were pulled. They averaged twenty-five and a half kilonewtons against thirty-one asked of them. So what was measured is about eight tenths of what is needed. That is short. It is also only two of them, and two is not enough to know how much the rest vary. None of that has to be settled before acting. Keep people out of the gym, test enough fixings to mean something, and design the repair.

TAKEAWAY 14W

Two tests can show something is short. They cannot show a building is safe.

4.3 Four hundred children on Monday

PUBLIC SAFETY · A ROOM · CHOICE

SCENE 37W

Four hundred children are due back Monday. The classrooms are sound and the gym is not. Bram Halvorsen, who runs the district's emergency work, has parents calling, and there is no spare school to move anybody to.

GUIDE 60W

All four options are fair readings of one site with two buildings on it. Ask of each whether it treats the site as one thing or as parts. The classrooms and the gym have different evidence behind them. Four hundred children arrive Monday and there is nowhere else. An answer that covers both buildings ignores half of what was measured.

ASK 9W

What should happen to Bay Road School on Monday?

BEHIND THE BUTTON 169W

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TAKES AS READ

a building can be used in part

VERDICT 79W

The fault is in one room, so the decision belongs to rooms rather than to the site. The classrooms have nothing wrong with them. The gym has heavy panels held on by fixings that tested below what the wall asks. If children can reach the classrooms without passing under those panels, the classrooms open and the gym stays shut. If the routes cross, that gets fixed first. Part-open is not optimism. It is a fence round the actual fault.

TAKEAWAY 16W

The question is which parts can be used, not whether the site is open or shut.

4.4 Holding it up, or hung on it

STRUCTURAL ASSESSMENT · A PERSON · BELT

SCENE 38W

Four hundred photographs came in this morning and the panel meets at four. Some of them show the parts that hold the building up. The rest show things the building holds, and both look bad in a photograph.

GUIDE 32W

Two bins. Ask whether the thing in the picture is holding something up. Columns, beams and walls that carry weight are. Ceilings, cladding, partitions and shelves are carried by the building instead.

ASK 12W

Send each photograph to the bin that says whether it carries weight.

BEHIND THE BUTTON 53W

Why it decides the placard. A cracked column changes whether the building stands up in the next shake. A cracked partition costs money and does not.

Why the other pile still matters. Falling cladding and ceilings hurt people. They need clearing and making safe, which is a different crew and a different day.

TAKES AS READ

some parts of a building hold it up and some are carried by it

VERDICT 145W

The question the panel has to answer is whether the building will stand up to the next shake, and only the parts that carry weight decide that. Columns, beams, load-bearing walls and the joints between them make a path from the roof down to the ground. Damage anywhere on that path makes the building weaker. A smashed partition, a ceiling on the floor, cladding hanging off the front and a toppled shelf are all carried by the building. None of them makes the frame weaker. That is why the worst-looking photographs are usually the ones that do not change the placard, and the photograph that does is often a thin crack on a column behind a wall that looks fine. The second pile still matters. Falling cladding hurts more people in a medium earthquake than anything else. It just wants a different crew, not shoring.

TAKEAWAY 16W

What decides if people can go in is what carries the weight, not what looks worst.

4.5 Ground that people put there

GEOTECHNICAL · A ROOM · CALLBACK · PROTOCOL

SCENE 37W

Elena Navarro, the ground engineer, has an 1892 survey and a photograph from 1948. Where Bay Road runs now there was a tidal creek. The port was pumped into place out of a dredger over four summers.

GUIDE 63W

Four kinds of ground on the left. Four ways ground behaves on the right. Pair them by asking two things. How loose is it? Can the water get out? Age is not the clue it looks like. One of these is the loosest and wettest of all. It is a line on an old survey, and you cannot see it from the street.

ASK 10W

Say what the ground under the Flats is made of

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

ground can be natural, or it can be put there by people · saturated means every gap between the grains is already full of water

VERDICT 80W

Fill that is pumped in through water lands loose, and it stays full of water. Shake it and the grains try to shuffle closer together. The water in the gaps cannot get out fast enough, so the water takes the load instead of the grains. Grains that are not pressing on each other cannot hold anything up, and the ground behaves like a thick liquid. Old ground has had a hundred centuries to pack down. This has had eighty years.

TAKEAWAY 12W

Loose, wet, young ground behaves in a way old rock does not.

END OF THE DAY 18W

The dangerous building is often the one nobody looked at, because nothing about it invited a second look.

DAY 5 OF 10

The hospital, and what waiting costs

BEFORE THE DAY — HUNT

Six props went out, and the log has four

Every temporary prop has to be found before the fence comes in. A prop nobody can find is either holding something up or lying in a street where a lorry will hit it. Four are where the log says.

PLAN CARD 77W

Wednesday, and the hospital has been yellow for four days. Peter Ives has ninety patients on the ground floor and two empty floors above them. Surgery is being done sixty kilometres away. Okonkwo wants the plant room opened and the walls cored before she signs. Bram Halvorsen says every day of that is a day of care nobody gets, and this morning he is right. Today you work out what is really unsettled, and what waiting costs.

BRIEFING 17W

90 patients, 3 days of yellow, and an argument about whether more evidence is worth more time.

WHAT YOU DECIDE 13W

Resolve a yellow placard on the one building the district cannot do without.

5.1 The part nobody has seen

STRUCTURAL ASSESSMENT · A ROOM · ATTEST

SCENE 36W

Ninety patients are still on the ground floor, and two floors above them stand empty. The hospital has been yellow for three days. The roof plant room is locked, with two full water tanks in it.

GUIDE 63W

Each claim on the list was signed or checked by somebody. Open the backing before you spend anything. Some claims rest on somebody looking at the thing they describe. Some rest on a drawing, an assumption, or a locked door. You get one visit. Hold the claims the evidence does not support, and spend the visit where being wrong would cost the most.

ASK 9W

Which claim still needs somebody to go and look?

BEHIND THE BUTTON 157W

What a signature does and does not say. It says somebody took responsibility for a sentence. It does not say anybody looked. Three days after an earthquake those two come apart all the time: drawings get checked, cladding gets looked at from the street, and a locked room gets written up from the last visit before the shake.

Why the tanks matter. Two full water tanks are a lot of weight, high up, in a room nobody has been into since the earthquake. If their supports have shifted, what happens next happens to the floor below — where ninety patients are. So that claim is the one that costs the most if it is wrong, whatever the paperwork says.

Why only one check. Every visit costs time the hospital does not have while it stays yellow. Choosing which claim to check is the real decision, and it is about consequence rather than about which record looks weakest.

TAKES AS READ

a building's parts can be cleared one at a time · a signature says somebody approved it, not that somebody looked

VERDICT 73W

A yellow placard should shrink to the specific things nobody has settled. The frame, the cladding and the stair have all been looked at, and there are notes to prove it. The roof plant room has not. Two full water tanks sit high in the building, and nobody has seen what holds them down. There is one visit left. Spending it on something already checked leaves the unchecked claim holding the review open.

TAKEAWAY 12W

A signed record is worth whatever the looking behind it was worth.

5.2 What four more days buys

HAZARD & FORECASTING · INCIDENT BASE · A PERSON · CHOICE

SCENE 33W

Tanaka has the aftershock rate for the week, and the hospital transfer log beside it. The office wants both stated as what they cost, rather than as principles. The board meets on Friday.

GUIDE 53W

All four options are ways of weighing four days. They differ in what they count. Ask of each whether it prices the waiting, the evidence, or neither. The frame has already been inspected. What nobody has entered is a room on the roof. So ask what four days of evidence would actually change.

ASK 15W

Four more days would sharpen the number. What does the waiting cost while it does?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

waiting has consequences, and they can be counted

VERDICT 79W

Both choices have a cost, and both can be counted. Waiting means transfers, delayed care and a hospital running short. Testing is worth it when it has a real chance of changing the answer. Here the frame has been looked at already, and the thing nobody has settled is a locked plant room with heavy tanks in it. Four days of coring answers a different question. Ask what the evidence could change, and who pays while it is collected.

TAKEAWAY 16W

Waiting is a decision, and it has to be compared with the decision to act now.

5.3 The instrument in the basement

SEISMIC NETWORK · A ROOM · VERIFY

SCENE 31W

There is an instrument in the hospital basement, fitted in 1998 and never needed until now. Cardoso has its record. It peaked at 0.31 g. The building was designed for 0.35.

GUIDE 57W

Say what the second instrument outside the hospital should read, then go and read it. Write the number down first. An instrument that has sat in a basement since 1998 has never been checked against anything, and the only way to find out whether it is telling the truth is another instrument that could disagree with it.

ASK 12W

Predict what the car park instrument read, then go and read it.

BEHIND THE BUTTON 139W

Why a second reading. One instrument on its own cannot be checked. If it is reading low, nothing in its own record says so. Another instrument nearby, on the same kind of ground, gives a number that either matches or does not.

What to expect. The car park instrument stands about eighty metres away on the same ground, so it should read close to the basement one — not exactly the same, because a basement is stiffer than open ground, but within about a fifth either way.

What 0.31 against 0.35 does not mean. The building was designed for 0.35, and coming in under a design figure is not a line below which nothing breaks. It says the shaking was about what the building was built for, which is worth knowing and is not a clean bill of health.

TAKES AS READ

an instrument inside a building measures the shaking that building got

VERDICT 148W

One instrument cannot check itself. The basement record says 0.31 g, and if that instrument has been drifting low since 1998 there is nothing in its own record to say so. The car park instrument stands eighty metres away on the same ground, and it reads 0.33 g. Those two agree, which is the useful result: the basement number can be trusted, and the hospital really did see about what it was designed for. Had the car park read 0.6, the basement instrument would have been the problem rather than the building. Two things the record still does not settle. Coming in under a design figure of 0.35 is not a line below which nothing breaks — it means the shaking was about the size the building was built for. And it says nothing about the plant room upstairs, which shakes differently from the basement it sits above.

TAKEAWAY 18W

A record of what happened is evidence, and it is never a line under which damage is impossible.

5.4 A notice somebody acts on

PUBLIC SAFETY · A ROOM · CALLBACK · SEQUENCE

SCENE 22W

Adeyemi has two hundred words, and a town that has heard two different numbers and a rumour about a bigger one coming.

GUIDE 60W

All four of these go in the notice. Nothing is being left out. Ask which one somebody can act on while standing in the street. Then ask which is a fact they will repeat later. Two hundred words are read in the order they are printed. A number going from 6.6 to 6.8 changes nothing about what to do now.

ASK 9W

Say it in a way that survives being repeated

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a public notice is read once, quickly, by somebody who is tired

VERDICT 71W

People act on the first thing they can use, so the instruction goes at the top. The reason comes next, because a reason is what makes an instruction survive being passed on. The revised number matters to scientists and changes nothing about what to do outside a building. The chance of aftershocks goes last, as a rate. As a rate it stays a chance. As a date it becomes a promise.

TAKEAWAY 18W

A notice earns its place by changing what somebody does, and everything else in it competes with that.

END OF THE DAY 17W

Gathering more evidence is not free, and evidence that cannot change the decision costs without buying anything.

DAY 6 OF 10

The material, and what it can still carry

PLAN CARD 74W

Friday, and the argument has moved from buildings to numbers. The library is a concrete frame from the 1970s, with cracks running diagonally through three ground-floor columns. Three engineers have called them moderate, serious and superficial. Duarte Ferreira points out that none of those is a number, and his lab can give one in a day. Today you turn a sample into a strength, and say what the building may be used for meanwhile.

BRIEFING 13W

The library's columns look cracked. Nobody has said what they can still hold.

WHAT YOU DECIDE 14W

Get a number for the strength of a damaged building instead of an opinion.

6.1 A number, with a method attached

MATERIALS & TESTING · A ROOM · CHOICE

SCENE 22W

Four cores, cut from the cracked columns. Sørensen crushes them: twenty-eight, thirty-one, twenty-four and thirty megapascals. The 1974 drawing asks for twenty-five.

GUIDE 56W

Four cores, and every option is a claim about what they settle. Ask of each how far it travels from four crushed cylinders. Strength is a fact about the concrete. What a column can carry also depends on the steel inside it and the load on it. The cores narrow the question rather than closing it.

ASK 21W

The drawing asks for twenty-five, and the lowest core came out at twenty-four. What does that tell you about the concrete?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

concrete strength is measured by crushing a sample of it

VERDICT 74W

The four cores average close to what the drawing asked for, so weak concrete is not the story. That is worth knowing, and it is not the answer. Crushing a cylinder measures the material. It does not measure what the column went through, how much steel is inside it, or how well that steel wraps the concrete. The cores narrow the question. Somebody still has to work out what the column can carry now.

TAKEAWAY 15W

A test tells you about the concrete. It does not tell you about the column.

6.2 Which way the cracks run

STRUCTURAL ASSESSMENT · A PERSON · CHOICE

SCENE 27W

The cracks run diagonally, both ways, crossing in an X across the middle of each column. They are fine, there are many, and they sit evenly apart.

GUIDE 58W

Four options, and the pattern has to be explained in full. The cracks go both ways. They cross. They are fine, many, and evenly spaced. Ask of each option what pattern it would leave instead. Crushing and settling do not change direction. And note the limit. A pattern names a cause, not what is left in the column.

ASK 16W

The cracks cross in an X on every column. What kind of movement makes that shape?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a crack opens across the direction something is being pulled apart

VERDICT 74W

Concrete cracks across whatever is pulling it apart. Push a column sideways and the pull runs on a diagonal, so the crack opens on a diagonal. Push it back and a second diagonal opens, crossing the first. That X is the mark of shaking, not of weight or of slow settling. Many fine cracks rather than one big one also says the steel inside is working. It does not say how much is left.

TAKEAWAY 14W

A pattern can point at the cause. It cannot tell you what is left.

6.3 Part of a building, for part of a purpose

PUBLIC SAFETY · A ROOM · SEQUENCE

SCENE 23W

Monday's public meeting has two hundred people booked. There is no other hall big enough. Three ground-floor columns in the library are cracked.

GUIDE 62W

All four of these will happen. The order is what is being decided. Ask of each what has to be settled before it means anything. A limit on numbers only works once you have said which part of the building. A review date is what stops a limit lasting for ever. Two hundred people are booked and there is no other hall.

ASK 9W

Decide what the building may be used for meanwhile

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

use can be limited by how many people, by which rooms, and for how long

VERDICT 62W

Part-use is an engineering line, not a hopeful label. Propping comes first, because it changes what the building can carry. A number on the door comes next. A limit nobody can count is not a limit. A review date comes after that, or the restriction quietly turns permanent and stops being believed. Only then is the meeting a yes or a no.

TAKEAWAY 20W

Restricted use is a real answer between open and shut, and it has to say what it is protecting against.

6.4 The same event, recorded twice

SEISMIC NETWORK · A ROOM · CALLBACK · BALLPARK

SCENE 33W

Inês Cardoso has two records of the same earthquake, from stations eleven hundred metres apart. One trace is short and sharp. The other is three times taller and runs half a minute longer.

GUIDE 56W

The same wave reached both stations. Work out how fast it travelled, from how far apart they are and how much later it arrived at the second one. Speed is the distance divided by the time. One tile is how much taller the second trace is, which is a different question about the same two records.

ASK 10W

How fast did the wave travel between the two stations?

BEHIND THE BUTTON 126W

Why the speed is worth knowing. It tells you what the wave travelled through. A wave through solid rock runs fast, around three or four kilometres a second. The same wave slows right down in loose wet ground, and that slowing is part of why it grows.

Why the two traces look so different. The far station stands on the Flats, where the ground is soft. Soft ground shakes further and for longer, so the trace is taller and runs on after the rock has gone quiet. Neither instrument is broken.

What each record is. A measurement of the shaking where that instrument stands. It is not a measurement of the earthquake, which is why two records of one earthquake can disagree and both be right.

TAKES AS READ

an instrument records the shaking where it stands, not the earthquake itself

VERDICT 141W

Both instruments caught the same wave. The far one caught it 0.4 of a second after the near one. Speed is distance ÷ time, so 1,100 ÷ 0.4 = 2,750 metres a second. That is fast, and it tells you what the wave came through: solid rock carries a wave at thousands of metres a second, and loose wet ground carries it far more slowly. The wave spent most of its journey in rock. The rest of the difference between the two records is about where the instruments stand. The far station stands on soft ground, so its trace is taller and it runs on for longer. Neither instrument is wrong. Each one recorded the shaking in its own place, which is what an instrument does, and it is why two records of one earthquake can disagree and both be right.

TAKEAWAY 17W

A record describes one place, so two records of one earthquake can disagree and both be right.

END OF THE DAY 21W

A word like serious is an opinion; a strength with a method behind it is something the next engineer can use.

The cordon nobody can lift

BEFORE THE DAY — CANVASS

Who slept in their own bed last night

The numbers the office is using came from Saturday, and they decide which buildings get checked first. Ask along the streets until enough answers have come back to say who went to the hall and who came home.

PLAN CARD 80W

Saturday, and the fence around the Flats is a week old. It went up on day one for falling masonry. It stayed on day three for the gas main. It is still up because nobody has said it can come down. Yvette Delacroix has four hundred families, sixty of them sleeping in the sports hall, and one question. What has to be true before people can go home? Today you find what each stretch of fence is really holding back.

BRIEFING 16W

11 streets have been closed for 6 days, and the reason is now 3 different reasons.

WHAT YOU DECIDE 13W

Weigh what a cordon protects against what it costs the people inside it.

7.1 Three reasons wearing one line

PUBLIC SAFETY · A ROOM · ATTEST

SCENE 32W

Shopkeepers wait at the tape with keys in their hands. Delacroix walks the line with the log. Three stretches have written reasons. The southern leg has none that anybody can point to.

GUIDE 71W

Each stretch of tape is a claim that there is a hazard there now. Open the evidence behind each one first. Some rest on somebody looking. Some rest on a hazard that has since been cleared. One has no written reason at all. You can go and check one stretch. Hold the claims the evidence does not support, and check the one where opening the wrong street would cost the most.

ASK 11W

Which closure most needs somebody to go and check the ground?

BEHIND THE BUTTON 152W

A closure is a claim. Tape goes up for a reason, and the reason is about a condition right now — falling glass, a loose parapet, ground that may still move. Conditions change and reasons expire, so a stretch nobody has been back to is a claim nobody has retested.

What a stale closure costs. Shopkeepers are at the tape with keys and deliveries. Every day a stretch stays shut with no hazard behind it, people trust the tape less — and then they walk around the stretches that do matter. Closing too much is not the safe mistake it looks like.

Why the unwritten one is not automatically the answer. A stretch with no logged reason may still be holding back something real that somebody saw on day one and never wrote down. Finding out is the point, which is why you spend the check instead of guessing from the log.

TAKES AS READ

a fence can be up for more than one reason · checking costs time, and there is one check left before the map is reissued

VERDICT 71W

A cordon usually holds several different hazards, and each one has its own condition for coming down. The Parade is closed for the masonry overhead. The gas leg waits on the utility. Ferry Street is about heavy vehicles getting through. The southern leg is different. The map keeps people out and the log gives no reason. One check is left, and the unbacked line is the one to spend it on.

TAKEAWAY 16W

Every stretch of fence is a claim about a hazard, and a claim should be checkable.

7.2 Soft ground, three days on

GEOTECHNICAL · A ROOM · CONTROL

SCENE 31W

Thandi Mbeki has been out on Ferry Street with a hand auger. The sand has dried and the water has dropped back. The surface carries a person, and not a truck.

GUIDE 65W

Four things are different about the street since Friday, and the plate test will give you a strength reading for whatever you set up. Change one of them and run the test. Change several and the reading cannot tell you which one did it. Then put your one back the way it was, because a change that does not undo itself was never the cause.

ASK 12W

Which difference is actually deciding how much weight Ferry Street will carry?

BEHIND THE BUTTON 139W

What went liquid and what came back. Shaking pushed the water between the sand grains until the grains stopped touching, and the ground behaved like a liquid. As the water pressure dropped the grains came back into contact, so the strength came back too.

What did not come back. The grains settled into new places, leaving the ground looser in some spots and hollow in others. That is why the surface carries a person and not a truck, and why it can go again if it is shaken while it is still wet.

Why one thing at a time. Water level, how packed the sand is, how heavy the load is and how long it sits are all different between Friday and today. A test that changes two of them at once gives a number that fits either story.

TAKES AS READ

ground that went liquid gets its strength back as the water pressure drops

VERDICT 157W

Four things about the street have changed since Friday. Only one of them moves the strength reading. Drop the water by three quarters of a metre, with everything else the same, and the plate test goes from 38 to 104. Let the water back up and it falls again. That is what makes it a real cause: the reading follows the change both ways. How tightly the sand is packed, and how long the load sits, move the reading less than the test's own wobble. This is the first day's mechanism run backwards. Water pressure between the grains is what let go. Taking the water away puts the grains back in touch. But the street is still weaker than it was before the earthquake, because the grains have settled somewhere new. So the honest answer is drained and disturbed. Walk on it. Keep trucks off it. Expect it to go again if it is shaken while wet.

TAKEAWAY 12W

The liquid part is over quickly. What it leaves behind is not.

7.3 A condition, not a date

HAZARD & FORECASTING · INCIDENT BASE · A ROOM · CHOICE

SCENE 29W

Halvorsen wants a date on the fence. Delacroix wants a condition instead. A date is a promise about the ground, and a condition is a promise about the office.

GUIDE 58W

Four options, and they differ in what the office would be promising. Ask of each whether the promise is about the ground or about the office's own work. A date promises something nobody controls. A rate is about the whole town, not one street. Residents want to see progress, and a written condition is progress they can check.

ASK 24W

A date is a promise about the ground, and a condition is a promise about the office. Which one can the office actually keep?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a promise can be tied to a day, or to something anybody can check

VERDICT 62W

A date promises how the ground will behave, and nobody controls that. A condition promises what the office will require before it moves a fence, and the office does control that. Publishing the conditions turns waiting into something people can watch. The aftershock rate is a poor trigger, because it says nothing about one street. Name the checks that open each stretch.

TAKEAWAY 17W

Tie a decision to something somebody can check, and it stays honest whatever the ground does next.

7.4 Keeping the water out of the trench

GEOTECHNICAL · A PERSON · HOLD

SCENE 33W

A trench has to stand open for an afternoon in ground that went liquid nine days ago. Pumps hold the water down. If the water comes back up, the sides do not wait.

GUIDE 47W

Hold the water level below the trench floor, inside the band on the gauge. The band gets narrower through the afternoon, because an open trench gets a little weaker every hour. Each thing that lets water in keeps letting it in. Set the pumps to match it.

ASK 12W

Hold the water below the trench floor while the shoring goes in.

BEHIND THE BUTTON 69W

Why the water matters. Sand holds together because the grains press on each other. Water between the grains pushes them apart, and then the sand behaves like a liquid.

Why rain is not one push. Rain keeps falling, a leaking pipe keeps leaking, and the tide keeps coming in. The water level does not settle at a new place. It keeps rising until the pumps match what is arriving.

TAKES AS READ

water between sand grains makes the ground weaker

VERDICT 123W

Sand holds together because the grains press on one another. Water between them pushes them apart, and once that happens the sand behaves like a liquid. That is exactly what happened to this ground nine days ago, in forty seconds. This afternoon it can happen slowly instead. Everything pushing the water up keeps pushing. Rain keeps falling, the leaking main keeps leaking, the tide keeps coming in. So the pumps are set to match what is arriving, and left there. A pump switched on for a minute buys a minute. The band gets narrower because an open trench loses a little strength every hour it stands. The same rise in water is a bigger problem at four o'clock than it was at one.

TAKEAWAY 14W

A leak that keeps arriving is not answered by turning a pump on once.

7.5 Where the first day of assessment goes

STRUCTURAL ASSESSMENT · A ROOM · CALLBACK · ALLOCATE

SCENE 34W

Six assessors are free for one day. Four hundred families are waiting on a decision in the Flats. Marina Court and the Parade are already fenced off, and Upper Town is mostly lived in.

GUIDE 65W

Six assessor-days is all you have, and each package you drag in spends some. Watch the list of questions rather than the buildings: it shows which ones your plan can answer. Every package you buy is a question you can answer and one you have given up. The fences already up cost nothing. Commit when the answers you can give are the ones today needs.

ASK 18W

Can your plan still get the waiting Flats families a decision today, knowing some questions have to wait?

BEHIND THE BUTTON 142W

Why the pool is the whole problem. Six assessors for one day is a fixed amount of attention, and the Flats alone could use all of it. The question is not what is worth checking — nearly everything is. It is which checks change what happens before tonight.

What makes an answer worth buying today. Four hundred families are waiting to hear whether they can go home. A check that can answer that by this evening is worth more than a better answer next week, because the delay has already made the decision for them.

Why the fences are on the board at zero. They are already up. They cost nothing to keep and they answer nothing new. They are there to be noticed: a plan that spends days re-checking what is already closed has bought a question it had already answered.

TAKES AS READ

an inspection queue is a decision, not a list

VERDICT 68W

There are six assessor-days and more good work than that. Every inspection bought is another one given up. The Flats sweep can send four hundred families home tonight, or keep them out on purpose. Marina Court and the Parade have fences doing that job already, so more detail there changes little today. Upper Town is useful and not urgent. Buy the answer that can still change what happens.

TAKEAWAY 14W

Spend a team on the questions whose answers can still change what happens today.

END OF THE DAY 22W

A cordon is a decision that is remade every day, and one kept out of habit is a decision nobody is making.

DAY 8 OF 10

The record that was wrong all along

BEFORE THE DAY — EVADE

The builder who wants a green notice in the car park

He has been outside since eight wanting somebody to say Marina Court is fine. Nothing is agreed anywhere except on the form. Keep clear of him and get your round done.

PLAN CARD 72W

Tuesday, and Inês Cardoso has found the thing nobody wanted found. The vault above Upper Town is the station every comparison this fortnight has been measured against. It is not founded on rock. It sits on four metres of broken granite that behaves like stiff soil, and it has been exaggerating the shaking since 1998. Today you work out what that does to the fortnight's conclusions, and which of them still stand.

BRIEFING 22W

The vault on the bench is not on rock. It is on four metres of weathered granite, and that changes every comparison.

WHAT YOU DECIDE 11W

Re-read the fortnight after finding the instrument everyone trusted was mis-sited.

8.1 Measuring against the wrong zero

SEISMIC NETWORK · A ROOM · TRACE

SCENE 25W

Cardoso's five-day check is finished. The station everything was compared against shakes more than it should. Navarro lays five of the fortnight's conclusions beside it.

GUIDE 59W

Open a conclusion and the panel shows what it was worked out from. Keep the ones that stand on a measurement of their own. Untick the ones whose evidence runs through the bad station. Then name the source the failing ones share. Both halves count: throwing away a good conclusion costs you as much as keeping a bad one.

ASK 12W

Which conclusions inherit the bad reference, and which stand on their own?

BEHIND THE BUTTON 154W

What went wrong. Every station's record is read against an assumed idea of how much that ground shakes. For one station that idea was wrong, and the five-day check has just proved it. Anything worked out through that station is wrong by the same amount. Anything measured somewhere else is not.

Why you should not throw out the lot. Withdrawing the whole fortnight looks careful. It is wrong twice. It throws away work that was never affected, and it tells the briefing that nothing is known when several things are. A retraction is a claim too, and it should be no bigger than the evidence.

How to read the chain. A conclusion is still good if there is some path to it that never touches the bad station. That is a question about the path, not about how big or alarming the number is — which is why you have to open each one.

TAKES AS READ

a comparison is only as good as the thing you compared against

VERDICT 77W

Every conclusion has a chain under it. The question is whether the bad station sits in that chain. The Flats figure was measured against that station, so it moves. The panel gives you both numbers you need. The station shakes 1.6 times what solid rock does, so 3.0 times the station is about 4.8 times rock. Anything copied from that figure moves as well. The hospital's own record and the Marina Court survey never touched the station.

TAKEAWAY 15W

A bad reference spoils whatever leaned on it, and leaves what was measured directly alone.

8.2 What still stands

GEOTECHNICAL · A PERSON · PROTOCOL

SCENE 28W

Navarro puts four of the fortnight's conclusions on the board and asks which of them leaned on the vault. The corrected vault report is clipped up beside them.

GUIDE 62W

Four conclusions on the left, and what the vault error does to each on the right. Trace each one back to its evidence. Then ask whether the vault is in that evidence. A survey and a field of sand do not need a reference station. Neither does an instrument in a basement. One bad comparison spoils what was compared, and nothing else.

ASK 11W

Match each conclusion to what the vault error does to it

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

a conclusion can rest on one measurement or on several

VERDICT 68W

Follow each conclusion down to what it rests on. The Flats figure used the vault as its yardstick, so a bad yardstick moves it. Marina Court was worked out from a survey, from heave and from sand. The hospital's number came from its own basement. The parapet case rests on what people watched fall. Those three may deserve other checks, and this error is not one of them.

TAKEAWAY 15W

A bad reference spoils what was measured against it, and leaves what was measured directly.

8.3 Five times, not three

HAZARD & FORECASTING · INCIDENT BASE · A ROOM · BALLPARK

SCENE 29W

If the Flats shake five times as hard as rock rather than three, a future earthquake delivers far more there. The district map is waiting on a new number.

GUIDE 54W

Two instruments recorded the same earthquake. One stood on the soft ground of the Flats, one on the rock above the town. How many times harder the Flats shook is the bigger reading divided by the smaller one. One tile is the old figure from the map, which is the number you are replacing.

ASK 11W

How many times harder did the Flats shake than the rock?

BEHIND THE BUTTON 119W

Why you divide. Asking how many times bigger one thing is than another is always a division. Both readings are in the same units, so the answer has no units at all — it is just a number of times.

What it means for the Flats. The rock and the soft ground got the same earthquake. The soft ground turned it into something much larger, and that is a fact about the ground rather than about the earthquake.

What the new number changes. Everything still to be built there, because the design figures come from the shaking that is expected. It changes nothing about the cracks already in the walls, which were made by shaking that has already happened.

TAKES AS READ

design figures come from the shaking that is expected

VERDICT 143W

Both instruments recorded the same earthquake. Written in per cent of gravity, the one on the Flats read 41 and the one on rock above the town read 8.2. How many times bigger is a division: $41 \div 8.2 = 5$. Both numbers are in the same units, so the answer is just a number of times, with no units on it. That is the whole finding, and it is about the ground and not about the earthquake — the same shaking arrived at both places, and the soft ground made it five times larger. The district map says three. Fixing it changes what gets designed for the Flats from now on, and the case for improving the ground there. It changes nothing that has already happened: the cracks in the walls were made by the real shaking, which the instruments measured directly.

TAKEAWAY 16W

Fixing what you expect changes what gets built next. It does not change what already happened.

8.4 What eight degrees does

STRUCTURAL ASSESSMENT · A ROOM · CALLBACK · CHOICE

SCENE 29W

Okonkwo has the column loads out. At eight degrees the weight above each column no longer lands over the middle of it. The gap grows the higher you go.

GUIDE 56W

All four options say something true about a leaning building. They differ in whether they name a force or a feeling. Ask of each whether an engineer could put a number on it. A column carries weight best straight down its middle. Eight degrees moves the weight off the middle, and the gap grows higher up.

ASK 10W

What is the engineering objection to leaving Marina Court standing?

BEHIND THE BUTTON 169W

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Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a column is built to be pushed straight down, along its own length

VERDICT 77W

A column is strongest when the weight lands straight down its middle. Lean the building and that weight arrives off to one side, so the column is bent as well as squashed. Bending is the job it was not built for, and the offset grows the further up the building you go. The question is not how frightening eight degrees looks. It is whether those columns have enough left in them for a load path nobody designed.

TAKEAWAY 18W

A column carrying its load off to one side is doing a job it was not built for.

END OF THE DAY 22W

A reference that is not what it claims moves every measurement made against it, and finding that out is a good day.

DAY 9 OF 10

Three things at once

PLAN CARD 62W

Wednesday, twenty to four, and three things arrive at once. A magnitude 5.1 aftershock has just shaken the town, the biggest in a week. A water main has burst on Ferry Street, into ground everybody now knows is loose. And the district wants a decision on the school gym by five. Okonkwo has four engineers. Today you decide what gets attention first.

BRIEFING 15W

An aftershock, a burst main and a deadline for the school, all inside an hour.

WHAT YOU DECIDE 9W

Rank three demands on one office in one afternoon.

9.1 Which of the three cannot wait

PUBLIC SAFETY · A ROOM · DELEGATE

SCENE 32W

Three calls land within minutes. A burst main is pouring water into the Flats. Buildings need checking after the 5.1. The school reopening message has to go out. Four engineers are free.

GUIDE 62W

Three calls, four engineers, and one problem you keep yourself. For each of the others, choose somebody who can take a real first action, and say what reading or event brings it back to you. An owner with no first action will come back for orders at the worst moment. Then take the watch on the one that needs your own judgement.

ASK 9W

How do you split the team across the three?

BEHIND THE BUTTON 146W

Which one to keep. One of these is getting worse while you decide. One could hurt somebody and needs checking. One is a deadline that is not going anywhere — the message will still need sending in an hour. Keeping the deadline is the usual mistake, because it is the one that finishes, and finishing it costs the field team its hours.

What a first action is. Not "keep an eye on it". Something the owner can do without asking you: shut the valve, take a reading at the boundary, put out the message. A handover made of instructions to think is not a handover.

Why every handover needs a way back. You are lending the problem, not giving it away. Saying which reading brings it back means somebody else can hold it without checking in, and you hear about it at the moment it matters.

TAKES AS READ

a problem still getting worse is different from one that has finished

VERDICT 72W

The burst main is still making the ground worse. So it gets stopped first, and stopping it belongs to the utility rather than to you. The 5.1 is over, and nobody knows what it did to the propped buildings. That is the judgement nobody else here can make, so you keep it. The school decision is settled and only needs saying. Three at once is not a queue. It is a split.

TAKEAWAY 19W

Keep the judgement nobody else can make, and hand the rest over with a first move and a number.

9.2 Wetting ground that has already failed

GEOTECHNICAL · A ROOM · CHOICE

SCENE 30W

Mbeki is on Ferry Street with a standpipe. The water under the road has risen three quarters of a metre since the main went. The surface is soft underfoot again.

GUIDE 53W

Four options for why a burst main matters more here. They differ in the reason. The pipes? The water pressure between the grains? Washing? The material itself? Ask of each whether it would also be true on granite. The leak on its own turns nothing to liquid. And the aftershocks have not finished.

ASK 14W

Why does the burst main matter more here than it would in Upper Town?

BEHIND THE BUTTON 169W

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TAKES AS READ

ground can only go liquid if it is full of water

VERDICT 65W

Ground holds things up because the grains press on each other. Water in the gaps carries some of the load instead. The more water there is, the less the grains carry. Loose wet fill is already near the edge, and a burst main pushes it closer before any shaking starts. The leak alone will not make it go liquid. It means a smaller shake will.

TAKEAWAY 13W

Wetting loose ground uses up the margin it had before the next shake.

9.3 A smaller shake against a weaker building

STRUCTURAL ASSESSMENT · A ROOM · BALLPARK

SCENE 30W

The 5.1 shook the Flats about a tenth as hard as the first earthquake. Eleven buildings have been propped since Friday, and two of them have new cracks this afternoon.

GUIDE 51W

Work out how much wall each metre of footing is holding up. Take the whole weight the wall puts down and share it along the length the footing runs. One tile is the height of the wall, which is part of why it is heavy and not part of this sum.

ASK 10W

How much weight is each metre of the footing holding?

BEHIND THE BUTTON 125W

Why per metre. A footing does not carry the weight in one place. It carries it spread along its whole length, so what matters is how much each metre has to hold, and that is the total shared out.

Why the new cracks matter so much. This shake was about a tenth as hard as the first one. A building that cracks under a tenth of what it already survived is not being hit harder — it is holding less.

What a prop does. It takes some of the load off, which is why the props went in on Friday. It does not put strength back into the wall, so the wall is still carrying what is left and still losing a little each time.

TAKES AS READ

a propped building is being held up, not repaired

VERDICT 144W

A footing carries its load spread along its length, so the number that matters is the load on each metre: total weight ÷ length. This wall puts 650 kilonewtons down along 12 metres of footing, which is 650 ÷ 12 ≈ 54 kilonewtons per metre. The wall was built to carry about 90, so on paper it has room. That is what makes the new cracks the important thing in the room. Today's shake was about a tenth as hard as the one on Friday, and a wall that cracks under a tenth of what it already stood up to is not being pushed harder — it is holding less than it was. The props help by taking some of the load off. They do not put strength back, so the wall keeps carrying what is left, and each shake takes a little more.

TAKEAWAY 16W

A small shake that still does damage is telling you how little the building has left.

9.4 What the teams are being sent to

PUBLIC SAFETY · A PERSON · SPOT

SCENE 28W

More buildings are waiting than the teams can reach. The office works to one priority. Marisol Okonkwo changes it as the fortnight goes on, and nobody announces it.

GUIDE 42W

Send teams to what the priority asks for and leave the rest waiting. The priority on the office board changes during the day. What counts is the buildings either side of a change, because they show whether you are reading the board.

ASK 13W

Send teams to the priority on the board, and keep checking the board.

BEHIND THE BUTTON 71W

Why the priority changes. It starts with buildings people are living in. After an aftershock it becomes anything already marked yellow, because those have less strength left. When the schools ask, it becomes schools.

Why the change is where it goes wrong. A team sent under the old priority spends half a day on a building nobody is waiting for, and the one that mattered waits another night with people inside.

TAKES AS READ

somebody decides which buildings are checked first

VERDICT 104W

Three priorities run across the fortnight. Buildings people are in, then buildings already marked yellow, then schools. A building can match two at once, and that is what makes a change cost something real. What the panel scores is the buildings either side of a change. Most of the queue is wanted by neither priority, so leaving those waiting proves nothing. The office's own version of this is the hour after an aftershock. The priority has changed to the yellow placards, because their strength has just dropped again, and the teams already in the vans are driving to the list from before the shake.

TAKEAWAY 18W

The cost of a rule that has changed is paid by whoever is still using the old one.

9.5 Why the number moved

HAZARD & FORECASTING · INCIDENT BASE · A ROOM · CALLBACK · CHOICE

SCENE 25W

Tanaka has both bulletins on the desk. The early one used the first stations to report. The later one used the whole network, hours afterwards.

GUIDE 53W

All four options explain a number that moved. They differ in what changed. The earthquake? The data? The sums? The scale? Only one of those can change six hours later. Ask what the second bulletin had that the first did not. Get it wrong and the town hears that the earthquake grew overnight.

ASK 18W

The second figure used more stations than the first. What got bigger — the earthquake, or the measurement?

BEHIND THE BUTTON 169W

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TAKES AS READ

a magnitude is worked out from records, so more records can change it

VERDICT 78W

A magnitude is not read off the ground. It is worked out from what the instruments recorded, so it gets better as more of them report. The first bulletin used the handful of stations that had answered in the first minutes. The later one used the whole network. Going from 6.6 to 6.8 sounds tiny, and it is not. That step means about twice the energy. The earthquake never changed. How much of it had been measured did.

TAKEAWAY 13W

A number that changes means the measurement improved. The event did not change.

END OF THE DAY 18W

Ranking is about which problem grows fastest and which is cheapest to remove, not about which sounds worst.

DAY 10 OF 10

What we know, and how well

BEFORE THE DAY — TAG

Catch Ives before the hospital handover

Ives, the hospital facilities manager, has the only plan with the yellow rooms marked on it, and he is walking back for the ten o'clock handover. After that he is on a ward round for four hours, and this afternoon's decision needs that drawing.

PLAN CARD 67W

Friday, two weeks on, and the fortnight goes into a report. Marisol Okonkwo will not sign one that states everything with the same confidence. A reader who cannot tell a measurement from a guess will act on both the same way. Rei Tanaka wants her forecast written as a range. Today you sort the findings by the evidence under them, and say what should outlive the emergency.

BRIEFING 20W

The report goes out today, and it has to say which of its findings are measurements and which are judgements.

WHAT YOU DECIDE 11W

State the fortnight's findings at the confidence each one has earned.

10.1 Measured, worked out, or guessed

SEISMIC NETWORK · A ROOM · CASEBOOK

SCENE 32W

Cardoso lays out four statements from the fortnight. The office believes all four. They are not believed for the same reasons, and she will not let them into the report as equals.

GUIDE 58W

Four statements the office believes, believed for four different reasons. Sort them by what somebody would have to produce to argue back. A second log? Another earthquake? A hole dug under the slab? Believing them is not the difference. The weakest of the four is the one the plan rests on, and the report has to say so.

ASK 6W

Sort the findings by their evidence

BEHIND THE BUTTON 153W

Why a clue needs an explanation and not a name. Saying what a clue is called is not saying what produced it. The board makes you commit to a cause, and a cause has to account for that clue and not for the one next to it. That is the difference between recognising something and explaining it.

How to use the one-each rule. Every explanation goes with exactly one clue. So the pairs you are sure about are worth more than themselves — each one you settle leaves fewer explanations for the clues you are unsure of. Two you trust can decide the other two.

Why one wrong line is impossible. If three lines are right, the last explanation has nowhere else to go, so it is right too. Being wrong always means two are wrong at once. When a join feels forced, look again at one you made early and stopped questioning.

TAKES AS READ

a claim can rest on a measurement, on reasoning, or on nothing much yet

VERDICT 70W

These four are believed for four different reasons, and the report should say which. The vault error was dug up, logged, and checked against another station for five days. The hospital's number came from its own basement, once. Marina Court's cause is good reasoning from a survey, with nobody having dug under the slab. The last one is a guess about the future. Saying so does not weaken the report.

TAKEAWAY 17W

Sorting findings by their evidence is what lets the next reader know which one to check first.

10.2 The number that has to be a range

HAZARD & FORECASTING · INCIDENT BASE · A PERSON · CHOICE

SCENE 30W

Tanaka's aftershock forecast goes into the report. She wants the range in the sentence. Okonkwo wants to know what the range is for. The staffing appendix waits on the wording.

GUIDE 58W

Four reasons to publish a range, and they differ in who it is for. Ask of each whether it is about being careful, being honest, being safe from blame, or planning. Nobody staffs an emergency for an average week. One number says nothing about how bad the bad week could be. The staffing appendix waits on this wording.

ASK 13W

Why does the forecast go in as a range rather than one number?

BEHIND THE BUTTON 169W

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TAKES AS READ

a forecast covers a spread of outcomes, not just one

VERDICT 72W

Nobody staffs an emergency for an average week. They staff it for a bad one, and how bad is exactly what the range says. Give a planner one number and they invent a margin themselves, usually too small. The range is also the honest half of the forecast. It says which part of the behaviour four hundred recorded aftershocks pin down well, and which part they do not. One number hides both.

TAKEAWAY 17W

A forecast with no range cannot size a decision, because the decision is sized against the range.

10.3 What outlives the emergency

STRUCTURAL ASSESSMENT · A ROOM · TRIAGE

SCENE 23W

Four things the office started doing under pressure. Okonkwo asks which is worth its cost on an ordinary Tuesday, when nothing has happened.

GUIDE 58W

All four practices are good ones. Two questions separate them. Does it stop the thing that actually went wrong? And will the office still do it in a quiet month? Trace each back to the mechanism. The gym, the roof room and the basement were all outside the first inspection. All three were written down, and nobody acted.

ASK 9W

Which practice would most directly stop this happening again?

BEHIND THE BUTTON 172W

Why first is a different question from best. When there is more than one call on the same hour, the value of doing something now is not how much it matters. It is how much is lost by doing it later. Something important that will be just as fixable in an hour can wait. Something small that will not be fixable can not.

What to look for in the options. Ask which one changes if you wait — a reading that is still moving, a person who is getting worse, a window that closes. Then ask which ones can only be done after something else. Those two questions usually leave one option standing.

Why only one answer is marked. Real shifts do all of these, in some order. The stop asks for the front of that order because that is the only part where the reasoning shows. Read the verdict afterwards even when you are right — it says what the others were waiting on, which is the rest of the answer.

TAKES AS READ

a habit taken up in an emergency has a cost that only shows later

VERDICT 67W

A habit worth keeping catches the mistakes that keep happening, at a price the office will still pay in a quiet year. The gym, the plant room and the Ferry Street basement were all outside what anybody looked at. In every case somebody knew. Writing that list down, and giving it an owner, would have caught all three. A rule nobody keeps is worse than no rule.

TAKEAWAY 20W

An unknown gets safer only when it has an owner, a date, and a rule that stops it being forgotten.

10.4 The slab is fine and the building is not

GEOTECHNICAL · A ROOM · CALLBACK · CHOICE

SCENE 35W

Navarro's survey: the slab is whole and level within itself, and the whole block has tipped. One side is down three hundred and forty millimetres, the other up ninety. Sand fans on the low side.

GUIDE 57W

Four options, and the survey has to be explained whole. The slab is not broken. It is level within itself. One side is down and the other is up. Sand came up on the low side. Ask of each option how many of those it covers. A frame that broke would bend, not tip in one piece.

ASK 19W

The slab is whole and level within itself, and the whole block has tipped. Where did the movement happen?

BEHIND THE BUTTON 169W

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TAKES AS READ

a foundation spreads a building's weight out into the ground

VERDICT 73W

Everything here points underneath the building rather than inside it. The slab is whole. The frame is straight. The whole block has tipped like a tray, one edge down and one edge up, with fresh sand pushed up on the low side. That is ground that stopped holding weight. Shake wet fill hard enough and the water takes the load, the grains stop pressing on each other, and one side sinks into it.

TAKEAWAY 10W

A building can fail without anything in the building breaking.

END OF THE DAY 21W

A finding is worth what its evidence is worth, and saying so is what makes a report usable a year later.

THE CLOSING CARD

The last cordon came down in the ninth week, street by street, each one when the hazard written against it had been dealt with rather than on a date anybody promised. The hospital never closed. Bay Road School lost its gymnasium for a term and kept its classrooms. Every parapet on the Parade is tied back now, which was done while the scaffolding was still up and cost a third of what it would have a year later.

What it cost: two weeks of a town living in halls, a six-storey block that will be taken down in the spring, and a recovery plan whose ground-improvement clause four hundred households have to find the money for. What is unfinished: the vault on the bench is still the reference station and still on weathered granite, the corrected amplification is a projection from nine boreholes, and the practice that would have caught all of it — writing down what an inspection could not see, and acting on that list — is a paragraph in a report until the next coordinator makes it a habit.

Card text only — no options, boards or answers. 8,307 words of card prose across 41 stops. Generated from `themes/aftershock_ms` by `tools/make-cardbook.mjs`.